

Marine Species Age Prediction: Crab vs Abalone Comparison

Mini Project Guide for Data Analysis & Machine Learning

Project Overview

Theme: Comparing age prediction in marine species using physical measurements **Goal:** Develop and compare machine learning models for predicting age in crabs and abalone using their physical characteristics

Dataset Details

Dataset 1: Abalone Age Prediction (UCI)

- **Source:** <https://archive.ics.uci.edu/dataset/1/abalone>
- **Size:** 4,177 instances, 8 features
- **Target:** Number of rings (age indicator)
- **Features:**
 - Sex (M/F/I - infant)
 - Length, Diameter, Height (mm)
 - Whole_weight, Shucked_weight, Viscera_weight, Shell_weight (grams)

Dataset 2: Crab Age Prediction (Kaggle)

- **Source:** <https://www.kaggle.com/datasets/sidhus/crab-age-prediction>
 - **Description:** Physical attributes of crabs found in Boston area
 - **Target:** Age (years)
 - **Features:** Similar physical measurements (length, diameter, height, weights, sex)
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Data Preprocessing Checklist

For Both Datasets:

- ☐ **Load and explore** basic dataset information
- ☐ **Check data types** and convert if necessary
- ☐ **Handle missing values** (check for NaN, empty strings, outliers)

- ☐ **Remove duplicates** if any exist
- ☐ **Categorical encoding:** Convert Sex/Gender to numerical (0/1 or one-hot)
- ☐ **Outlier detection:** Use IQR or Z-score methods
- ☐ **Feature scaling:** StandardScaler or MinMaxScaler for ML models

Expected Data Quality Issues:

- Missing or inconsistent values in weight measurements
 - Outliers in physical dimensions
 - Categorical variables needing encoding
 - Potential measurement unit inconsistencies
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Exploratory Data Analysis (EDA) Plan

1. Basic Statistics

```
python  
  
# For both datasets  
df.info()  
df.describe()  
df.head()
```

2. Missing Data Analysis

```
python  
  
# Check missing values  
df.isnull().sum()  
# Visualize missing data pattern
```

3. Correlation Analysis

- Create correlation heatmaps for both datasets
- Compare which features correlate most with age
- Identify multicollinearity issues

4. Visualizations to Create

- **Distribution plots:** Age distribution in both species
- **Box plots:** Physical measurements by sex/gender

- **Scatter plots:** Weight vs Age, Length vs Age
 - **Pair plots:** Relationships between all numerical features
 - **Count plots:** Sex/Gender distribution
 - **Histograms:** Each numerical feature
 - **Comparison plots:** Side-by-side comparisons of similar features
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Machine Learning Pipeline

Required Models (Minimum):

1. **Linear Regression** - For continuous age prediction
2. **Logistic Regression** - Convert age to categories (young/adult/old)

Optional Additional Models:

- Random Forest Regressor
- Support Vector Machine (SVM)
- Decision Tree
- XGBoost (advanced)

Model Evaluation:

- **Regression metrics:** MSE, RMSE, MAE, R^2
 - **Classification metrics:** Accuracy, Confusion Matrix, Precision, Recall
 - **Cross-validation:** K-fold validation for robust results
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Report Structure Guide

1. Introduction (1-2 pages)

- Problem statement: Why predict marine animal age?
- Dataset selection rationale
- Project objectives and expected outcomes

2. Dataset Description (1-2 pages)

- **Abalone Dataset:** Features, size, biological significance
- **Crab Dataset:** Features, size, comparison with abalone

- **Data quality assessment:** Initial observations

3. Data Preprocessing (2-3 pages)

- Missing value handling strategies
- Duplicate removal process
- Categorical encoding methods
- Outlier detection and treatment
- Feature scaling/normalization

4. Exploratory Data Analysis (3-4 pages)

- **Correlation analysis:** Feature relationships in both datasets
- **Visualization insights:** Key patterns discovered
- **Species comparison:** How crab vs abalone features differ
- **Age distribution:** Patterns in both species

5. Machine Learning Models (2-3 pages)

- Model selection rationale
- Training and validation process
- Performance comparison between datasets
- Feature importance analysis

6. Results & Comparison (2-3 pages)

- **Model performance:** Accuracy/metrics for both datasets
- **Species comparison:** Which species' age is easier to predict?
- **Feature analysis:** Most important features for each species
- **Insights:** Biological interpretations

7. Conclusion (1 page)

- Key findings summary
- Project limitations
- Future improvement suggestions

Presentation Structure (5 minutes)

Slide 1: Title & Team Introduction

- Project title, team members, datasets chosen

Slide 2: Problem Statement

- Why predict marine animal age?
- Brief dataset overview

Slide 3: Data Overview & Preprocessing

- Dataset sizes and features
- Key preprocessing steps taken

Slide 4: Key Visualizations

- 2-3 most interesting plots
- Correlation insights

Slide 5: Machine Learning Results

- Model performance comparison
- Best performing models for each dataset

Slide 6: Species Comparison & Insights

- Which species' age is easier to predict?
- Key biological insights

Slide 7: Conclusion & Questions

- Summary of findings
- Open for questions

Creative Comparison Ideas

Biological Insights:

- **Growth patterns:** How do physical traits change with age in each species?
- **Predictive features:** Which measurements are most important for each species?
- **Species efficiency:** Which species shows clearer age-measurement relationships?

Data Science Insights:

- **Data quality:** Which dataset required more preprocessing?
- **Model performance:** Which dataset is easier to predict?
- **Feature engineering:** What transformations worked best for each?

Visualization Comparisons:

- Side-by-side correlation heatmaps
 - Overlaid age distribution plots
 - Feature importance comparison charts
 - Prediction accuracy comparison
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File Organization

```
project_folder/  
├── data/  
│   ├── abalone.data  
│   ├── crab_age.csv  
│   └── processed/  
├── notebooks/  
│   ├── 1_data_exploration.ipynb  
│   ├── 2_preprocessing.ipynb  
│   ├── 3_eda_visualization.ipynb  
│   ├── 4_machine_learning.ipynb  
│   └── group_number.ipynb (final combined)  
├── reports/  
│   ├── final_report.pdf  
│   └── presentation.pptx  
└── visualizations/  
    ├── correlation_plots/  
    ├── distribution_plots/  
    └── comparison_plots/
```

Getting Started Steps

1. Download datasets:

- Abalone: From UCI repository

- Crab: From Kaggle (requires free account)

2. **Quick exploration:**

- Load both datasets
- Check shapes, features, data types
- Identify immediate preprocessing needs

3. **Register group:** Fill the Google Sheet with your team details

4. **Start coding:** Begin with data loading and basic exploration

Important Reminders

- **No code screenshots** in the report
 - **One submission** per group
 - **Print one report copy** for presentation
 - **5-minute presentation** maximum
 - **Viva questions** will be asked about your code
 - **Be creative** and surprise your instructor!
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Good luck with your marine species age prediction project! 🐚 🦀